

Code:

```
#include <stdio.h>

long long a[3005];
int F[3005];
int G[3005];

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int i, j;
    for (i = 0; i < n; i++) {
        scanf("%lld", &a[i]);
    }

    // Compute F(i): next greater element index
    for(i=0;i<n;i++) {
        F[i] = -1;
        for (j = i + 1; j < n; j++) {
            if(a[j]>a[i]) {
                F[i] = j;
                break;
            }
        }
    }

    // Compute G(i): next smaller element index
    for (i = 0; i < n; i++) {
        G[i] = -1;
        for (j = i + 1; j < n; j++) {
            if (a[j] < a[i]) {
                G[i] = j;
                break;
            }
        }
    }

    // Output A[G(F(i))] or -1
    for (i = 0; i < n; i++) {
        if (F[i] != -1 && G[F[i]] != -1) {
```

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        printf("%lld", a[G[F[i]]]);  
    } else {  
        printf("-1");  
    }  
    if (i < n - 1) printf(" ");  
}  
printf("\n");  
  
return 0;  
}
```

Input:

```
8  
3 7 1 7 8 4 5 2
```

Output:

```
1 4 4 4 -1 2 -1 -1
```